

RAAKESH MENON A

SENIOR DATA ENGINEER | LEAD DATA ENGINEER

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PROFESSIONAL SUMMARY

Senior Data Engineer with 4+ years of experience delivering enterprise data platforms and analytics solutions for global clients across banking, semiconductor manufacturing, aerospace manufacturing, and animal health. Strong in Python, Advanced SQL, Snowflake, PySpark, Apache Spark, ETL modernization, data migration, data modeling, distributed processing, data quality, validation, anomaly detection, and Power BI. Experienced in modernizing legacy pipelines, building scalable cloud data workflows, optimizing query and pipeline performance, and supporting end to end Agile delivery. Hands on with Microsoft Azure data engineering and DevOps, with solution evaluation across Azure Data Factory, Azure Data Lake Storage Gen2, Azure SQL, Azure Functions, Azure Synapse Analytics, Azure Databricks, Delta Lake, Microsoft Fabric, and OneLake.

CORE TECHNICAL SKILLS

Programming and Data Processing: Python, Advanced SQL, PL SQL, Perl, Pandas, NumPy, PySpark, Apache Spark, Shell Scripting, Unix Linux, Regex, REST API Integration, JSON, XML

Data Engineering: Snowflake, ETL Development, ETL Modernization, Data Pipeline Architecture, Scalable Data Pipelines, Data Migration, Data Transformation, Data Integration, Data Warehousing, Data Modeling, Distributed Data Processing, Workflow Orchestration, Backend Data Engineering, Big Data Processing, Structured and Semi Structured Data Processing, Talend, Batch Data Processing, Incremental Data Processing, Source to Target Mapping, Reusable ETL Components

Data Quality and Optimization: Data Quality Engineering, Data Validation, Data Profiling, Data Reconciliation, Source to Target Validation, Anomaly Detection, Query Optimization, Performance Optimization, Statistical Analysis, Predictive Analytics, Reusable Validation Frameworks, Data Quality Controls, Validation Automation

Azure and Cloud: Microsoft Azure, Azure Data Factory, Azure Data Lake Storage, ADLS Gen2, Azure SQL, Azure Functions, Azure Synapse Analytics, Azure Databricks, Delta Lake, Microsoft Fabric, OneLake, Azure DevOps, Azure Boards, Azure Repos, Azure Pipelines, Cloud Data Engineering, Cloud Data Migration, Cloud Data Integration, Cloud Solution Evaluation

Machine Learning and AI: Anomaly Detection, Time Series Forecasting, Regression, Random Forest, Gradient Boosting, XGBoost, Feature Engineering, Model Evaluation, MLflow, Experiment Tracking, Model Registry, MLOps, Generative AI, LLM Integration, Prompt Engineering, Retrieval Augmented Generation, LLM Assisted Data Engineering

BI and Engineering Delivery: Power BI, DAX, Power Query, Tableau, Looker Studio, Advanced Excel, KPI Dashboard Development, Enterprise Analytics, Reporting Automation, Git, CI CD, Agile Methodology, SDLC, Release Coordination, Deployment Coordination

PROFESSIONAL EXPERIENCE

Senior Systems Engineer / Senior Data Engineer

Apr 2023 – Present

Tata Consultancy Services (TCS) — Bengaluru, India

- PROJECT 6 —Semiconductor Manufacturing Data Engineering: Led enterprise analytics and data engineering for approximately 2 TB of manufacturing data across 300+ BOM and product hierarchy schemas, using Python, Advanced SQL, Snowflake, PySpark, Power BI, and scalable ETL to integrate SAP and Teamcenter engineering data.
- Automated validation, reconciliation, profiling, anomaly detection, statistical analysis, predictive analytics, KPI scoring, and reporting workflows, reducing validation and reconciliation effort by 95% and supporting KPI dashboards for approximately 250 users.
- Improved pipeline and reporting performance by 95% through SQL and Snowflake optimization, scalable transformation workflows, backend processing, data modeling, and Azure DevOps based CI CD and deployment coordination.
- PROJECT 5 — Financial Services Data Engineering: Designed an enterprise data migration and analytics engineering solution aligned to large scale financial data workflows, using Python, Advanced SQL, PySpark, Apache Spark, Snowflake, ETL, distributed processing, data modeling, and Power BI to process approximately 500K records across 5 source systems.
- Built scalable migration and transformation pipelines with 20+ Spark and ETL workflows, applying structured and semi structured data processing, workflow orchestration, query optimization, and data quality controls to improve processing performance by approximately 95%.
- Enabled enterprise analytics and reporting for approximately 150 to 200 consumers through curated Snowflake datasets, optimized data models, Power BI reporting, and reusable data engineering components.

- PROJECT 4 — Financial Services | Team 2 | Enterprise Data Migration and Analytics Engineering: Designed Apache Spark and PySpark pipelines for approximately 500K records across 10 source systems, supporting distributed processing, scalable ETL, migration, transformation, integration, and Snowflake warehousing.
- Improved pipeline performance by 95% through distributed execution tuning, query optimization, scalable transformation architecture, data modeling, and structured and semi structured data processing.
- Enabled enterprise analytics workflows for approximately 150 to 200 reporting consumers through curated Snowflake datasets, Power BI dashboards, enterprise reporting, workflow orchestration, and Agile delivery.
- AZURE SOLUTION EVALUATION: Evaluated Azure Data Factory, ADLS Gen2, Azure SQL, Azure Functions, Azure Synapse Analytics, Azure Databricks, Delta Lake, Microsoft Fabric, and OneLake against ingestion, transformation, orchestration, serving, scalability, performance, and integration requirements when assessing cloud data engineering solutions.
- AZURE ARCHITECTURE ASSESSMENT: Compared service fit, workload patterns, data volume, transformation complexity, latency, operational overhead, security, CI CD, and downstream analytics needs to recommend appropriate Azure components for migration, ETL, distributed processing, data warehousing, and reporting solutions.

Systems Engineer / Data Engineer

Dec 2021 – Mar 2023

Tata Consultancy Services (TCS) — Bengaluru, India

- PROJECT 3 — Financial Services | Team 1 | ETL Modernization and Automation: Modernized 60+ legacy ETL workflows by transitioning Perl based processing patterns to reusable Python architectures integrated with Talend, SQL, Snowflake, and enterprise automation frameworks.
- Reduced processing time by 90% through ETL optimization, dependency reduction, reusable transformation components, query optimization, and workflow automation, while developing 50+ reusable components for scalable data engineering delivery.
- Accelerated release and engineering delivery by 100% through reusable automation, CI CD enabled workflows, Git based development, SDLC practices, release coordination, and AI assisted development for code generation, validation, and optimization.
- PROJECT 2 — Aerospace Manufacturing Data Engineering: Engineered scalable Python and SQL ETL workflows processing approximately 100K manufacturing records across 5+ source systems, supporting data transformation, integration, modeling, warehousing, and enterprise analytics.
- Improved reporting performance by 85% through query optimization, ETL optimization, scalable backend processing, workflow automation, and structured and semi structured data transformation using Snowflake, PySpark, Python, and Advanced SQL.
- Eliminated identified data inconsistencies across 20+ ETL workflows through standardized transformation, source integration, data modeling, validation controls, and BI enablement across Power BI, Tableau, and Looker Studio.
- PROJECT 1 — Animal Health Data Engineering: Developed Python based ETL pipelines and backend data processing for enterprise healthcare datasets, processing approximately 10 million records across 20+ ETL workflows using Python, Advanced SQL, Snowflake, reusable ETL components, and workflow orchestration.
- Reduced a manual process from 48 hours to approximately 2 hours, an improvement of up to 95.83%, through Python automation, SQL validation, reusable transformation components, and scalable Snowflake processing.
- Improved processing performance by 80% by optimizing SQL workflows, pipeline execution, source to target validation, structured and semi structured data integration, and reusable data engineering frameworks.
- AZURE SOLUTION EVALUATION: Evaluated Azure Data Factory, ADLS Gen2, Azure SQL, Azure Functions, Azure Synapse Analytics, Azure Databricks, Delta Lake, Microsoft Fabric, and OneLake as solution options for enterprise ETL, cloud migration, data integration, distributed processing, and analytics workloads.
- AZURE DATA PLATFORM ASSESSMENT: Assessed ingestion, orchestration, transformation, storage, serving, monitoring, scalability, performance, and downstream BI requirements to determine suitable Azure architecture patterns and integration approaches for enterprise data pipelines.

ACHIEVEMENTS

- Earned multiple stakeholder appreciation awards from stakeholders and internal leadership for delivering 10+ high impact data engineering and analytics solutions improving operational efficiency and reporting reliability.
- Conducted 10+ technical knowledge sharing and upskilling sessions on Python, SQL, and ETL best practices for 25+ engineers, strengthening team capability in analytics engineering.

EDUCATION

Bachelor of Engineering | Anna University, Chennai, India

2016 – 2020

CERTIFICATIONS

- Google Advanced Data Analytics Professional Certificate
- IBM Data Science Professional Certificate
- Google Business Intelligence Professional Certificate
- Generative AI with Large Language Models | DeepLearning.AI